

SC Honors Geometry - Nappier
Homework 10.2 – Finding Arc Measures

Name: _____

Find each arc measure.

1. AB is a diameter of circle P and $m\widehat{AC} = 54^\circ$. Find $m\widehat{CB}$.
2. $m\widehat{XY} = 113^\circ$. Find the major arc $m\widehat{XZY}$.
3. What does every semicircle measure, and how do you name one?

Circle graphs.

4. A category takes up 30% of a circle graph. Find its central angle.
5. A survey shows 45% dogs, 20% birds, 35% cats. Find all three central angles.

Arc addition.

6. Three arcs make a full circle. Two measure 102° and 134° . Find the third.
7. $\widehat{AB} = 47^\circ$ and $\widehat{BC} = 56^\circ$ are adjacent. Find $m\widehat{ABC}$.

Congruent and similar arcs.

8. Two arcs both measure 83° , in two circles of different sizes. Congruent or similar? Explain.
9. When are two circles congruent? When are they similar?
10. How many degrees does the minute hand of a clock sweep in 20 minutes?